

Carmel Holy Word Secondary School  
2024-2025 Term 2  
Secondary 3 Computer Literacy  
Practical Assessment  
Time: 45 Minutes

|           |      |
|-----------|------|
| Name      |      |
| Class     |      |
| Class no. |      |
| Date      |      |
| Marks     | / 50 |

1. Write your Name, Class and Class Number in the boxes provided.
  2. Follow the instructions below to complete the examination tasks.
  3. **If you fail to upload your work, NO MARK will be given.**
- 

1. **ALL TASKS** should be finished in **VS Code**.
2. Login to Teams and open the assignments named “S3 Term 2 Practical Assessment – Task 1”, “S3 Term 2 Practical Assessment – Task 2”, “S3 Term 2 Practical Assessment – Task 3” and “S3 Term 2 Practical Assessment – Task 4”.
3. Download the files “task2\_2425.py”, “task3\_2425.py” and “task4\_2425.py” to your computer.
4. Submit your work to the appropriate assignments in Teams.

## Task 1

(10 marks)

Write a program to calculate and display the Body Mass Index (BMI), as shown in Figure 1 for Task 1.

- (a) **Create** a Python file in VS Code and **name** it “task1\_2425.py”.
- (b) **Ask** the user to input their weight in kilograms.
- (c) **Ask** the user to input their height in meters.
- (d) **Calculate** the BMI using the formula:  $BMI = weight / (height ** 2)$ .
- (e) **Print** the calculated BMI. (as shown in **Figure 1**).
- (f) **Save the file as** “task1\_2425.py” and **submit** it to Teams.

```
Enter your weight in kg: 65
Enter your height in meters: 1.7
Your BMI is: 22.49134948096886
```

Figure 1

## Task 2

(10 marks)

Write a program to input multiple math scores and calculate the highest, lowest, and average scores, as shown in Figure 2 for Task 2.

- (a) **Download** “task2\_2425.py” from Teams and **open** it in VS Code.
- (b) **Create** an empty list to store the scores.
- (c) **Ask** the user to input math scores.
- (d) **Convert** the input string into an **integer**.
- (e) **Complete** the if-elif-else condition.
  - If the input number is -1, stop the loop
  - If the input number is greater than 0, add the number to the list
  - If the input number is neither -1 nor greater than 0, print out the number and inform the user that the number is lower than 0 (as shown in **Figure 2**).
- (f) **Print** the highest score, lowest score, and average score. (As **Figure 2** has shown).
- (g) **Save the file as** “task2\_2425.py” and **submit** it to Teams

```
Enter a number (or -1 to stop): 45
Enter a number (or -1 to stop): 12
Enter a number (or -1 to stop): 0
Enter a number (or -1 to stop): -3
-3 is lower than 0.
Enter a number (or -1 to stop): -1
The maximum score is: 45
The minimum score is: 0
The average score is: 19.0
```

Figure 2

### Task 3

(15 marks)

Write a program to convert text to speech, as shown in Figure 3 for Task 3.

- (a) **Download** “task3\_2425.py” from Teams and **open** it in VS Code.
- (b) **Import** the Text-to-Speech and playsound libraries.
- (c) **Set up** the text-to-speech engine.
- (d) **Ask** the user for a sentence to convert to speech.
- (e) **Save** the audio output as an mp3 file.
- (f) **Play** the mp3 file.
- (g) **Print** a confirmation message after saving.
- (h) **Save the file as** “task3\_2425.py” and **submit** it to Teams.

```
Please enter a sentence to convert to speech: Jesus Love You
Jesus Love You has been converted and played.
```

Figure 3

#### Task 4

(15 marks)

Write a program to read a video and detect faces, as shown in Figure 4 for Task 4.

- (a) **Download** “task4\_2425.py” from Teams and **open** it in VS Code.
- (b) **Import** the OpenCV library
- (c) **Open** the video file “video.mp4”.
- (d) **Read** frames from the video.
- (e) **Convert** the frame to grayscale
- (f) **Use a rectangle** to mark detected faces in each frame.
- (g) **Display** the video with marked faces.
- (h) **Save the file as** “task4\_2425.py” and **submit** it to Teams.

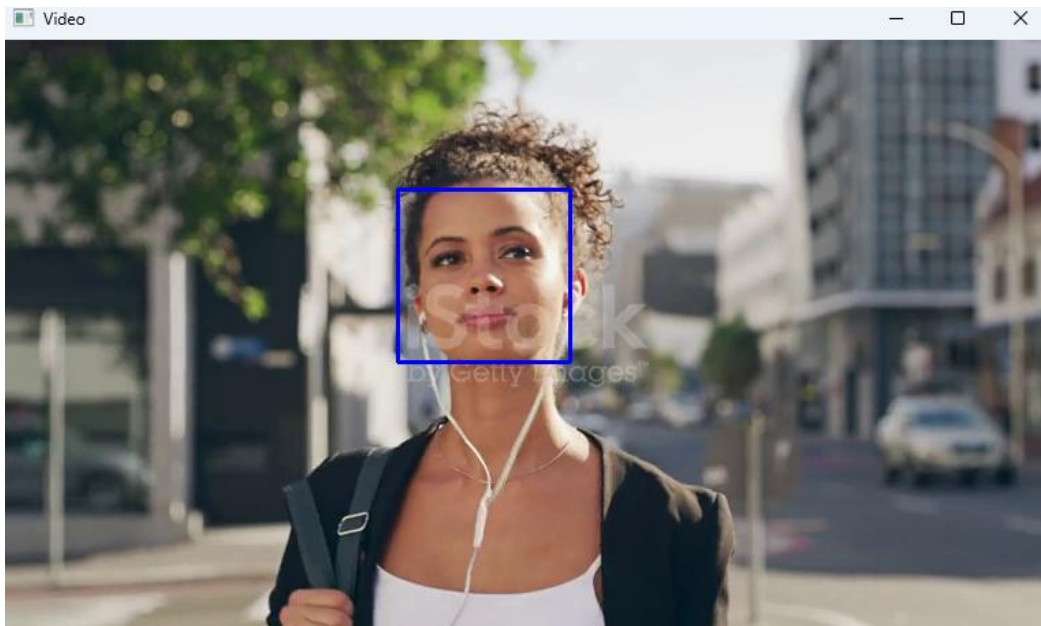


Figure 4

**End of Paper**

## Appendix: Modules and Functions API Reference Table

| Module                 | pip install                                | Example of Import Statement                                     |
|------------------------|--------------------------------------------|-----------------------------------------------------------------|
| gTTS                   | gTTS                                       | <code>from gtts import gTTS</code>                              |
| Playsound              | <code>playsound==1.2.2</code>              | <code>from playsound import playsound</code>                    |
| Google Cloud Translate | <code>google-cloud-translate==2.0.1</code> | <code>from google.cloud import translate_v2 as translate</code> |
| Speech Recognition     | SpeechRecognition                          | <code>import speech_recognition as sr</code>                    |
| OpenCV                 | <code>opencv-contrib-python</code>         | <code>import cv2</code>                                         |

| Module                 | Description                            | Function Example                                                                                                             |
|------------------------|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| gTTS                   | Setup gTTS                             | <code>audio = gTTS(text="I Love You", lang="en")</code>                                                                      |
| gTTS                   | Save TTS audio to a file               | <code>audio.save("audio.mp3")</code>                                                                                         |
| Playsound              | Play the audio file                    | <code>playsound("audio.mp3")</code>                                                                                          |
| Google Cloud Translate | Initialize the translation client      | <code>client = translate.Client()</code>                                                                                     |
| Google Cloud Translate | Translate text between languages       | <code>result = client.translate(text, source_language="en", target_language="zh")<br/>print(result["translatedText"])</code> |
| OpenCV                 | Read an image file                     | <code>image = cv2.imread("face.jpg")</code>                                                                                  |
| OpenCV                 | Open a video file or camera stream     | <code>cap = cv2.VideoCapture("video.mp4")</code>                                                                             |
| OpenCV                 | Read a frame from the video            | <code>isTrue, frame = cap.read()</code>                                                                                      |
| OpenCV                 | Convert the frame to grayscale         | <code>cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)</code>                                                                         |
| OpenCV                 | Draw a rectangle around detected faces | <code>cv2.rectangle(frame, (x, y), (x + w, y + h), (255, 0, 0), 2)</code>                                                    |
| OpenCV                 | Display the video frame in a window    | <code>cv2.imshow('Video', frame)</code>                                                                                      |
| OpenCV                 | Wait for a key event (with delay)      | <code>cv2.waitKey(1)</code>                                                                                                  |